

RGB COLOUR DETECTOR

Using TCS3200 Sensor Module

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This is a simple colour sensor using Arduino Uno R3 and TCS3200 colour sensor module. It can be useful for colour identification and detection for food-processing units, colour printer applications, paint-mixing applications and other industrial applications including robotics.

This project is used for detecting primary colours (red, green and blue, or RGB)—colours that are physically available in LEDs in one package; for example, common cathode or common-cathode RGB LED. We can display primary colours and also generate specific colours by modifying the Arduino code. The project demonstrates the basic interfacing of

TCS3200 sensor, Arduino Uno and common-cathode RGB LED.

TCS3200 colour sensor module (SEN0101) is shown in Fig. 1 and microscopic view of the RGB arrays is shown in Fig. 2. On the microscopic level, you can see the square boxes inside the eye on the sensor. These square boxes are arrays of the RGB matrix. Each of these boxes contains three sensors: one each for sensing red light, green light and blue light intensity. It is better than TCS230 colour sensor module. This sensor can be used to identify any number of colours with an accurate programming code.

Circuit and working

Fig. 3 shows the circuit diagram of the RGB colour detector using TCS3200. It works off 9V power supply connected across connector CON1. However, an Arduino Uno board requires only 5V. So it has a bridge rectifier with a regulator that converts 9V to 5V logic, which can further be converted to 3.3V with the help of LM1117 voltage regulator.

Brain of the circuit is Arduino Uno R3 board having ATmega328 or ATmega328P microcontroller (MCU). It has 14 digital input/output (I/O) pins and six analogue input pins, 32k flash memory, 16MHz crystal oscillator,

PARTS LIST	
<i>Semiconductors:</i>	
Board1	- Arduino Uno R3
RGB1	- Common-cathode RGB LED
TCS3200	- Colour sensor module
<i>Resistors (all 1/4-watt, $\pm 5\%$ carbon):</i>	
R1	- 100-ohm
<i>Miscellaneous:</i>	
CON1	- 2-pin connector
CON2	- 11-pin connector
	- 9V battery

USB connection, power jack, ICSP header and reset button.

TCS3200 module has eight pins as shown in Fig. 4. This module consists of programmable colour light-to-frequency converters that combine configurable silicon photodiodes and current-to-frequency converter on a single monolithic CMOS integrated circuit. Output is square-wave (50 per cent duty cycle) with frequency directly proportional to light intensity (irradiance).

Digital inputs and outputs allow direct interface to the MCU or other logic circuitry. Output enable (OE) places the output in high-impedance state for multiple units sharing an MCU input line. In TCS3200, the light-to-frequency converter reads an 8×8 array of photodiodes. Sixteen photodiodes have blue filters, another sixteen have green, yet another sixteen have red and remaining sixteen are clear with no filters.

All photodiodes of the same colour are connected in parallel. Pins S2 and S3 of TCS3200 are used to select the group of photodiodes (red, green, blue and clear) that are active. The detailed pin description is shown in Tables I, II and III, respectively.

Each sensor array in these three



Fig. 1: TCS3200 colour sensor module



Fig. 2: Microscopic view of TCS3200 chip

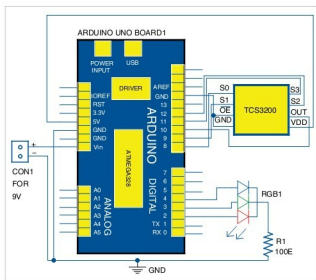


Fig. 3: Circuit diagram of the RGB colour detector using TCS3200